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* Friday office hours

HW Talk:
- always use what's given before

ex

Lemma 5 $\forall m, n \in \mathbb{N}$,

$$m + \sigma(n) = \sigma(m+n)$$

thinking: $m + \sigma(n) = \sigma^{\sigma(n)}(m)$ — by def. +
$\sigma(m+n) = \sigma(\sigma^n(m))$

Axiom 5: $\sigma^{\sigma(n)} = \underbrace{\sigma \circ \sigma^n}_{\sigma(\sigma^n(\cdot))}$

Pf:

For $m, n \in \mathbb{N}$, consider $m + \sigma(n)$.
By def. of addition, we have

$$m + \sigma(n) = \sigma^{\sigma(n)}(m).$$

By Iteration axiom $\sigma^{\sigma(n)}(m) = \sigma(\sigma^n(m))$.

By def. of (+), $\sigma(\sigma^n(m)) = \sigma(n+m)$.

We conclude $m + \sigma(n) = \sigma(m+n)$ $\square$

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$1+(2+3) = (1+2)+3$ ✓ Associative law

$\sigma^2(1) = \sigma \circ \sigma(1) = \sigma(2) = 3$

Lemma 8:
If $n \in \mathbb{N}$, then $\sigma^{(n)}(0) = n$. In particular $0+n = n$.

Lemma 9:
If $n \in \mathbb{N}$, then $1+n = \sigma(n)$

Thm. 10:
If $n \in \mathbb{N}$, then $x+y = y+x$

Pf. sketch: Use induction (Axiom 4)

Let $S_x = \{n \in \mathbb{N} : x+n = n+x\}$

Then, Lemma 8 gives the base case:
$x+0 = x = 0+x$ ✓ $0 \in S_x$ ✓

thm. 4 lemma 8

Suppose $j \in S$

$x+\sigma(j) = \sigma(j)+x$ using Lemma 9 & induction hyp.

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§ 1.5 : Multiplication

* Multiplication is iterated addition. *

<u>Def. of multiplication:</u>

$x \mapsto x+m$ "maps to"

Let $m, n \in \mathbb{N}$. define

$$\alpha_m : \mathbb{N} \to \mathbb{N} \text{ by } \alpha_m(x) = x+m$$

As usual, we write α_m^n for the nth iterate of α_m.

We define $m(n) \overset{\text{def.}}{=} \alpha_m^n(0)$

<u>ex</u>

Compute $2 \cdot 3$.

Let $m=2$, $n=3$.

$2(3) = \alpha_3^2(0) = 6$

Since $n=3$, we must take the 3rd iterate of 2.

$\alpha_3^2(0) = (((0+2)+2)+2)) = 6$

PROF WORK:

$2 \cdot 3 = \alpha_2^3(0) \cancel{= \alpha_2^{\alpha(2)}(0) = \alpha_2 \circ \alpha_2^2(0)}$

PROF WORK:

$$2\cdot 3 = \alpha_2^3(0) = \alpha_2^{\sigma(2)}(0) = \alpha_2 \circ \alpha_2^2(0)$$

$$= \alpha_2(2+2) \qquad = \alpha_2 \circ \alpha_2 \circ \alpha_2^1(0)$$

$$= \alpha_2(2+\sigma(1)) \qquad = \alpha_2 \circ \alpha_2 \circ \alpha_2(0)$$

$$= \alpha_2(\sigma^{\sigma(1)}(2)) \qquad = \alpha_2 \circ \alpha_2 \circ (0+2)$$

$$= \alpha_2(\sigma \circ \sigma^1(2)) \qquad = \alpha_2 \circ \alpha_2(2) \text{ — Lemma 8}$$

$$= \alpha_2(\sigma \circ \sigma(2))$$

$$= \alpha_2(\sigma(3)) = \alpha_2(4) = 4+2$$

$$= \sigma^2(4)$$

$$= \sigma^{\sigma(1)}4$$

$$= \sigma \circ \sigma^{(1)}4$$

$$= \sigma \circ \sigma(4)$$

$$= \sigma(5) = 16.$$

<u>Thm. 11</u> $\forall m \in \mathbb{N},\ m \cdot 0 = 0$

<u>Pf</u>:

We compute $m \cdot 0 = \alpha_m^0(0)$ by def. of mult.

Now, α_m^0 is the identity function by Ax. 5.

So $\alpha_m^0(0) = id(0) = 0$.

Thus $m \cdot 0 = 0$.

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Lemma 12: $\forall m, n \in \mathbb{N}$, $m(\sigma(n)) = (m \cdot n) + m$.

Thm. 13: $\forall m \in \mathbb{N}$, $m \cdot 1 = m$.

Thm. 14: $\forall x, y, z \in \mathbb{N}$, $(x+y)z = xz + yz$

- right distributivity

Pf sketch:

Fix x, y. Induct on z.

$S_{x,y} = \{z \in \mathbb{N} : (x+y) \cdot z = xz + yz\}$

Lemma 15: If $n \in \mathbb{N}$, $0 \cdot n = 0$.

Lemma 16: If $n \in \mathbb{N}$, $1 \cdot n = n$.

Thm. 17: (Commutativity of mult.)

$\forall x, y \in \mathbb{N}$, $x \cdot y = y \cdot x$

Pf:

Let $x \in \mathbb{N}$. Defining $S_x = \{y \in \mathbb{N} : xy = yx\}$
We proceed w/ induction.

We have $x \cdot 0 = 0$ by Thm. 11. & $0 \cdot x = 0$
by Lemma 15

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Pf:

Let $x \in \mathbb{N}$. Defining $S_x = \{y \in \mathbb{N} : xy = yx\}$ we proceed w/ induction.

We have $x \cdot 0 = 0$ by Thm. 11. & $0 \cdot x = 0$ by Lemma 15.

Thus $x \cdot 0 = 0 \cdot x$. $0 \in S$. (base case)

Now suppose $n \in S_x$. (induction hyp) We compute

w+s $x \cdot \sigma n = x \cdot n + x$ (by Lemma 12.)

$= n \cdot x + x$ (by hyp)

$= n \cdot x + 1 \cdot x$ (by Lemma 16)

$= (n+1)x$ (thm. 14)

$x \cdot \sigma n = \sigma n \cdot x$ (thm. 6)

Thus $x \cdot \sigma n = \sigma n \cdot x$.

so $\sigma(n) \in S$.

Since $0 \in S_x$ & $\sigma(n) \in S_x$ whenever $n \in S_x$, Axiom 4 shows $S_x = \mathbb{N}$.

$S_x = \{y \in \mathbb{N} : xy = yx\} = \mathbb{N}$

Since x was arbitrary, we conclude $\forall\, xy = yx \in \mathbb{N}$.

$\square$